

Object Oriented Programming Lab Record

Experiment No: 07

Title: Implementation of different types of inheritance, function overriding, and virtual functions in C++.

Objective:

To understand and implement various types of inheritance and demonstrate function overriding and the use of virtual functions in C++.

Theory:

1. Inheritance: It allows a derived class to inherit properties and behaviors from a base class.

Types of Inheritance:

- Single Inheritance
- Multiple Inheritance
- Multilevel Inheritance
- Hierarchical Inheritance
- Hybrid Inheritance

2. Function Overriding: When a derived class defines a function with the same name and signature as in its base class.

3. Virtual Function: A base class function declared with the keyword 'virtual' that can be overridden in the derived class.

Program:

```
#include <iostream>
using namespace std;

class Base {
public:
    virtual void show() {
        cout << "Base class show() function" << endl;
    }
};

class Derived1 : public Base {
public:
    void show() override {
        cout << "Derived1 class show() function" << endl;
    }
};

class Base2 {
public:
    void display() {
        cout << "Base2 class display() function" << endl;
    }
};

class Derived2 : public Derived1, public Base2 {
public:
    void show() override {
        cout << "Derived2 class show() function" << endl;
    }
};

class Derived3 : public Derived2 {
```

```

public:
    void show() override {
        cout << "Derived3 class show() function" << endl;
    }
};

int main() {
    Base *b;
    Derived3 d3;
    b = &d3;
    b->show();

    Derived2 d2;
    d2.display();
    d2.show();

    return 0;
}

```

Output:

Derived3 class show() function
 Base2 class display() function
 Derived2 class show() function

Result:

Different types of inheritance, function overriding, and the concept of virtual functions were successfully implemented and demonstrated in C++.

Viva Questions and Answers:

1. Q: What is inheritance in C++?

A: Inheritance is the mechanism by which one class acquires the properties and behaviors of another class.

2. Q: What is the difference between function overloading and function overriding?

A: Overloading occurs in the same class, while overriding occurs in derived classes with the same function signature.

3. Q: What is a virtual function?

A: A virtual function allows runtime polymorphism by enabling derived classes to override base class methods.

4. Q: What is runtime polymorphism?

A: Runtime polymorphism means that function calls are resolved at runtime using virtual functions.

5. Q: What are the advantages of inheritance?

A: It promotes code reusability, reduces redundancy, and improves maintainability.